

Møbee Desktop Layout Strategy – Authoritative Specification

Purpose

This document defines a robust, device-independent layout strategy for the Møbee web game. Its goal is to eliminate card overflow, inconsistent vertical positioning, and screen-size-dependent behavior across all desktop displays (MacBook Pro, 27" iMac, etc.) at default 100% browser zoom.

This is not an implementation guide yet. It is a conceptual and architectural specification intended for Claude to implement precisely and minimally.

Problem Summary

The current layout relies on fixed-size (600×600) boards combined with JavaScript-based transform scaling using guessed header/footer heights (e.g. 80px). This causes:

- Over-scaling on large displays (27" iMac), leading to cards flowing off-screen
- Under-utilization of space on other desktops
- Vertical misalignment between screens (main page, gameplay, friends lobby)
- Layout behavior that changes with browser zoom level (175% “fixes” issues accidentally)

The root cause is that transform scaling does not participate in layout and is computed from window dimensions instead of real available space.

Core Principle (Single Source of Truth)

There must be exactly one authoritative concept for card sizing and positioning:

The stage (cards area) must be sized from the actual remaining container space, not from the window and not from guessed header heights.

The remaining container already exists in the DOM: the flex area below the header (e.g. #board-container). This container represents the true usable space.

Target Mental Model

1. The page is vertically divided into: • Header (logo, timer, score, UI chrome) • Stage container (everything below the header)
2. The stage container owns sizing authority.
3. The stage itself is always a perfect square: **stageSize = min(container width, container height)**
4. All card layouts (gameplay, main menu, friends lobby) are rendered inside this stage using the same sizing logic.

Non-Negotiable Rules

- Do NOT compute scale from `window.innerHeight` minus guessed constants
- Do NOT use transform-based scaling as the primary sizing mechanism
- Do NOT use asymmetric padding to “nudge” layouts into place
- Do NOT allow different screens to invent their own sizing logic

Unified Stage Strategy

There must be a single sizing pipeline applied consistently:

1. Measure the real bounding box of the stage container
2. Compute `stageSize = min(width, height)`
3. Apply `stageSize` as a real width/height (or via a CSS variable)
4. Center the stage using flexbox
5. Render cards relative to the stage, not the viewport

This pipeline must be used identically for: • Main page cards • Gameplay cards • Play-with-friends lobby cards

Vertical Alignment Requirements

- The stage must be visually centered within the available container
- Any spacing above or below the stage must be intentional and symmetric
- Header height changes must automatically reflow the stage without code changes
- No screen should appear “too high” or “too low” relative to others

Zoom Independence

At 100% browser zoom on any desktop: • No card may overflow the viewport • No card may be clipped • The stage must appear maximized but stable

Zooming the browser must not be required to achieve correct layout.

Optional (Product-Level, Not Required)

A maximum stage size clamp may be applied to avoid excessively large boards on ultra-wide displays, but this is optional and should be explicit and deliberate.

Acceptance Criteria

- Layout is stable on MacBook Pro and 27" iMac at 100% zoom
- No transform-based scaling artifacts remain
- All screens share the same visual center logic
- Changing header height does not break card placement
- The implementation is minimal, readable, and unified

Implementation Guidance

Claude should:

- Identify and remove window-based scale math

- Use the stage container as the sizing authority
- Apply one consistent sizing model across all screens
- Avoid screen-specific hacks or magic numbers

This document is the authoritative reference.